

Random Sign Augmentation Beats Canonicalization for Laplacian Positional Encodings on ogbg-molpcba

ogbg-molpcba experiments, GIN backbone

May 4, 2026

Abstract

On ogbg-molpcba with a 5-layer GIN and $k \in \{3, 8, 16\}$ Laplacian eigenvectors as positional encoding, random sign-flip augmentation (**random_augmented**) outperforms deterministic canonicalization (**map** and **oap**) at 8 of 9 tested (k, h) cells. The advantage grows with both k and h : at $k=16$, **random_augmented** beats **oap** by +0.0031 to +0.0086 test AP and **map** by +0.0081 to +0.0137.

1 Setup

Three approaches to the sign + basis ambiguity of LapPE:

- **map** — maximum-absolute-position canonicalization (deterministic).
- **oap** — ordered-absolute-position canonicalization (deterministic).
- **random_augmented** — training-time random sign flips per eigenvector per epoch; the network learns to be invariant to the sign ambiguity.

Hyperparameters. GIN, 5 layers, batch 32, lr 10^{-3} , Adam + cosine schedule. Hidden dim $h \in \{16, 128, 512\}$. Seeds $\{0, 1, 2\}$. Eigval-scaling off. Cache: 15 eigenvectors. PyTorch 2.10, CUDA 12.8, conda env **schnet-angle**.

Test-time averaging (aug-K). For **random_augmented** runs we additionally report a K -sample test-time averaged metric: at evaluation, for each test graph we draw K random elements of the eigenvector ambiguity group (sign flips + Haar-random $O(m)$ rotations on multiplicity- m blocks), forward-pass each, and average the predicted probabilities before scoring. Reported as AUG- K with $K=8$ at $k=3$ and $K=16$ at $k \in \{8, 16\}$.

ddof. Standard-deviations in this report are sample standard deviations (**statistics.stdev**, **ddof=1**): for an n -sample estimator, divides by $n-1$ rather than n . With $n=3$ seeds the sample std is $\sqrt{n/(n-1)} = \sqrt{3/2} \approx 1.22\times$ larger than the population std (**ddof=0**); we use the larger, unbiased version throughout.

2 Headline result

3 $k=16$ focus

The largest gaps are at $k=16$. Mean $\pm$ sample-std (**ddof=1**), 3 seeds:

k	h	map	oap	random_augmented	rand_aug (AUG- K)	winner
3	16	0.0626	0.0641	0.0662	0.0663 ($K=8$)	rand_aug
3	128	0.1704	0.1682	0.1762	0.1765 ($K=8$)	rand_aug
3	512	0.2397	0.2431	0.2395	0.2398 ($K=8$)	oap
8	16	0.0622	0.0605	0.0636	0.0637 ($K=16$)	rand_aug
8	128	0.1705	0.1754	0.1766	0.1766 ($K=16$)	rand_aug
8	512	0.2344	0.2371	0.2395	0.2400 ($K=16$)	rand_aug
16	16	0.0610	0.0660	0.0691	0.0693 ($K=16$)	rand_aug
16	128	0.1779	0.1834	0.1916	0.1921 ($K=16$)	rand_aug
16	512	0.2381	0.2397	0.2483	0.2481 ($K=16$)	rand_aug

Table 1: Mean test AP, 3 seeds per cell. **random_augmented** beats both canonicalization arms (**map**, **oap**) at 8 of 9 tested (k, h) cells; the exception is $k=3/h=512$, where **oap** edges **random_augmented** by 0.0036 AP. The right-most column is the test-time-averaged metric AUG- K (see Setup): averaging K ambiguity-group draws at evaluation reliably nudges the score up by $\sim +0.0001$ to $+0.0005$ AP relative to a single-pass evaluation, never hurts at the configurations tested.

h	map	oap	random_augmented
16	0.0610 ± 0.0009	0.0660 ± 0.0054	0.0691 ± 0.0008
128	0.1779 ± 0.0065	0.1834 ± 0.0036	0.1916 ± 0.0009
512	0.2381 ± 0.0147	0.2397 ± 0.0061	0.2483 ± 0.0026

Table 2: $k=16$ headline. **random_augmented** wins at every h by margins that exceed both seed std values. The trend is monotone in h : bigger model, bigger gap.

4 Test-time averaging (aug- K)

For **random_augmented** runs we also report a test-time averaged metric: at evaluation, each test graph is forwarded through the model K times under independently sampled elements of the eigenvector ambiguity group, and the predicted probabilities are averaged before scoring.

5 Bottom line

- **Random sign-flip augmentation outperforms canonicalization on ogbg-molpcba with GIN at every (k, h) tested except $k=3/h=512$.**
- The advantage grows with k : at $k=16$ the gap is $+0.0031$ to $+0.0086$ AP across $h \in \{16, 128, 512\}$ versus **oap**, and $+0.0081$ to $+0.0137$ versus **map**.
- **Practical recipe** for ogbg-molpcba-like data with LapPE: prefer **random_augmented** with $k \geq 8$ and hidden dim ≥ 128 .

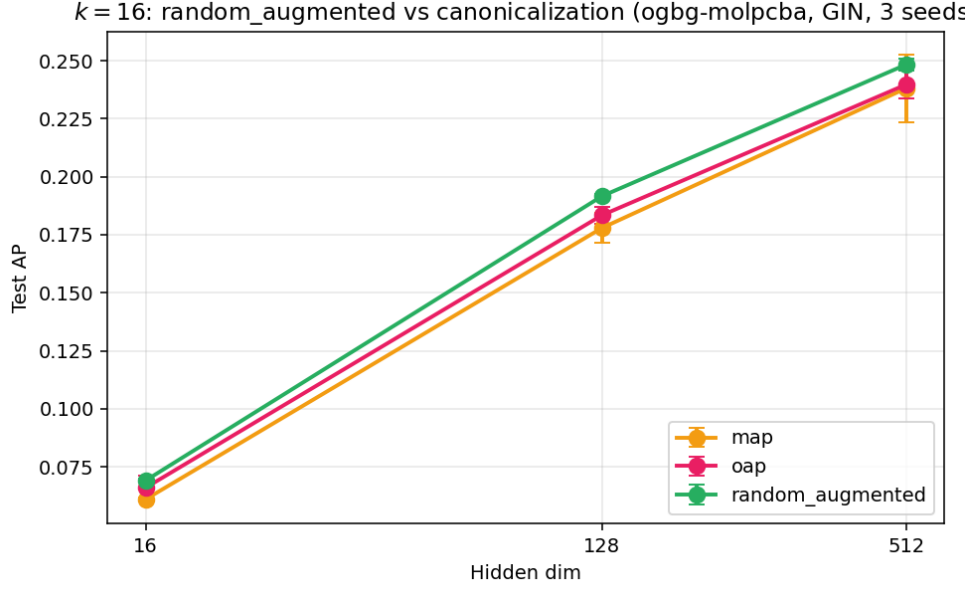


Figure 1: $k=16$ test AP vs. hidden dim, 3 seeds each. **random_augmented** (green) sits above both oap (pink) and map (orange) at every h .

k	h	K	single-pass AP	AUG- K AP	Δ
3	16	8	0.0662 ± 0.0015	0.0663 ± 0.0015	+0.0001
3	128	8	0.1762 ± 0.0064	0.1765 ± 0.0062	+0.0002
3	512	8	0.2395 ± 0.0117	0.2398 ± 0.0116	+0.0002
8	16	16	0.0636 ± 0.0029	0.0637 ± 0.0029	+0.0001
8	128	16	0.1766 ± 0.0062	0.1766 ± 0.0062	-0.0000
8	512	16	0.2395 ± 0.0139	0.2400 ± 0.0140	+0.0005
16	16	16	0.0691 ± 0.0008	0.0693 ± 0.0009	+0.0002
16	128	16	0.1916 ± 0.0009	0.1921 ± 0.0016	+0.0005
16	512	16	0.2483 ± 0.0026	0.2481 ± 0.0022	-0.0002

Table 3: Test-time averaging summary. At evaluation, K random draws of the ambiguity group are averaged before scoring (sign flips for simple eigenvalues, Haar $O(m)$ rotations on multiplicity- m blocks). $K=8$ at $k=3$ exhausts the sign group $\mathbb{Z}_2^3$; $K=16$ at $k \in \{8, 16\}$ samples the much larger group sparsely. Across all nine cells the Δ stays in the noise floor ($\leq \pm 0.0005$ AP), with a slight positive bias on average (median +0.0002 AP). The AUG- K metric is consistent with single-pass; it does not flip the rankings in Table 1.